

Chapter 1 Continuous Function on $\mathbb{R}$, Limits of Function

1.1 Sequence

Topology of $\mathbb{N}$ given by $(\mathbb{N}, <)$.

1.1.1 Cluster Points

Definition 1.1.1.1 Let $(a_n) \in \text{Func}(\mathbb{N}, \mathbb{R})$ or $\text{Func}(\mathbb{N}, \mathbb{C})$. z is a **cluster point** of (a_n) if

$$\forall \epsilon > 0 \forall N_0 \exists n > N_0 \wedge a_n \in B_\epsilon(z)$$

Equivalently, for all $B_\epsilon(z) \cap \{a_n | n \geq N_0\} \neq \emptyset, \forall N_0$.

1.1.2 Convergence

(Definition in general in topology)

Definition 1.1.2.1 Given $(a_n) \in \text{Func}(\mathbb{N}, \mathbb{R})$ or $\text{Func}(\mathbb{N}, \mathbb{C})$. Then

$$a_n \rightarrow z := (\forall \epsilon \exists N_0 (A_{N_0} \subseteq B_\epsilon(z)))$$

where $A_{N_0} := \{a_n | n \geq N_0\}$.

Equivalently, $\forall \epsilon \exists N_0 (n \geq N_0 \rightarrow |a_n - z| < \epsilon)$. Take the notion of

$$\lim_{n \rightarrow \infty} a_n = z$$

(In this section) $P((a_n), A) := \forall \epsilon \exists N_0 (n > N_0 \rightarrow a_n \in B_\epsilon(A))$

◇ **Lemma 1.1.2.2** (Uniqueness of Convergence in $\mathbb{R}, \mathbb{C}$) If $P((a_n), A), P(a_n, B)$, then $A = B$.

◦ **Proof** Via Hausdorffness. □

Operations

♠ **Proposition 1.1.2.3** (Operations limits of $\mathbb{R}$ or $\mathbb{C}$) Suppose the following limits exists:

$$\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} (a_n \cdot b_n) = (\lim_{n \rightarrow \infty} a_n) \cdot (\lim_{n \rightarrow \infty} b_n)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}, \lim_{n \rightarrow \infty} b_n \neq 0$$

◦ **Proof**

- Suppose $A = \lim_{n \rightarrow \infty} a_n, B = \lim_{n \rightarrow \infty} b_n$. Now selection $N = \max\{N_1, N_2\}$, $|a_n - A| < \epsilon/2, |b_n - B| < \epsilon/2$, then $|(a_n + b_n) - (A + B)| < \epsilon$. With uniqueness.

- Similarly.
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□

◇ **Lemma 1.1.2.4** (Majorance of Sequence) Suppose $(a_n), (b_n), (c_n)$ be sequence. Then, we have

$$a_n \leq b_n \leq c_n, n \geq N_0 \Rightarrow \lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n \leq \lim_{n \rightarrow \infty} c_n$$

◦ **Proof** It is easy to verify: Suppose $a_n \leq b_n, n \geq N_0$. If $\lim_{n \rightarrow \infty} a_n > \lim_{n \rightarrow \infty} b_n$, then

$$b_n < B + \epsilon < A - \epsilon < a_n, \epsilon = (B - A)/2$$

Contradiction.

□

◇ **Lemma 1.1.2.5** (The Sequence with Limit is Bounded) Suppose $\lim_{n \rightarrow \infty} a_n = A$, then $\{a_n\}$ is bounded.

◦ **Proof** Omitted.

□

Average

♠ **Proposition 1.1.2.6** Suppose $\lim_{n \rightarrow \infty} a_n = A$, then

$$\lim_{n \rightarrow \infty} \frac{a_0 + \dots + a_n}{n+1} = A$$

◦ **Proof** Suppose $n > N_1, |a_n - A| < \epsilon$. Now, select some N_2 such that

$$\frac{(a_0 - A) + \dots + (a_{N_1} - A)}{n+1} < \epsilon, n > N_2$$

so, $N = \max\{N_1, N_2\}$,

$$\left| \frac{(a_0 - A) + \dots + (a_N - A)}{N+1} \right| \leq \left| \frac{a_0 + \dots + a_{N_1}}{N+1} \right| + \left| \frac{a_{N_1+1} + \dots}{N+1} \right| < 2\epsilon$$

□

1.1.3 Monotone Sequence

Definition 1.1.3.1 Let (a_n) be a sequence

- (a_n) is **strictly increasing** if $a_{n+1} > a_n$;
- (a_n) is **increasing** if $a_{n+1} \geq a_n$.

Similarly for decreasing.

♠ **Proposition 1.1.3.2** If (a_n) is increasing and has a upper bound, then

$$\lim_{n \rightarrow \infty} a_n = \sup\{a_n | n \in \mathbb{N}\} \in \mathbb{R}$$

◦ **Proof** Firstly the limit exists: Because of completeness of $\mathbb{R}$, $\sup\{a_n | n \in \mathbb{N}\} \in \mathbb{R}$. Then omitted. $\square$

1.1.4 Subsequence

Definition 1.1.4.1

- We firstly define the **strictly increasing function** of index

$$\sigma : \mathbb{N} \rightarrow \mathbb{N}$$

such that: $\sigma(n) < \sigma(n+1), n \in \mathbb{N}$.

- Let $(a_n) \in \text{Func}(\mathbb{N}, S)$ a subsequence is $(a_{\sigma(n)})$.

► **Theorem** (Bolzano-Weistrasss) Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

◦ **Proof** Furthermore, this theorem has a equivalent explanation: $\mathbb{R}$ is secondly countable, and there exists at least one cluster point.

Such point could be given by

$$\sup\{a_n | n \in \mathbb{N}\}$$

And the elements by countable basis. $\square$

Furthermore, we could extend ∞ to the topology by compactification, so bounded condition could be removed.

Definition 1.1.4.2 Define:

- $\overline{\lim} a_n := \lim_{N \rightarrow \infty} \sup_{n \leq N} a_n$;
- $\underline{\lim} a_n := \lim_{N \rightarrow \infty} \inf_{n \in \mathbb{N}} a_n$;

♠ **Proposition 1.1.4.3** We have

$$\lim_{n \rightarrow \infty} \sup(a_n) \geq \lim_{n \rightarrow \infty} \inf(a_n)$$

◦ **Proof** Omitted. $\square$

1.1.5 Cauchy Sequence

Definition 1.1.5.1 A sequence (a_n) is **Cauchy sequence** if

$$\forall \epsilon \exists N (N \in \mathbb{N} \wedge \forall n \forall m (n > N \wedge m \geq 0 \rightarrow |a_{m+n} - a_n| < \epsilon))$$

♠ **Proposition 1.1.5.2**

Cauchy Sequence $\Leftrightarrow$ Convergent.

◦ **Proof**

- $\Rightarrow$ If the sequence is Cauchy, then take monotone $\overline{\lim} a_n$ and $\underline{\lim} a_n$. We have
- $\Leftarrow |a_{n+m} - a_n| \leq |a_{n+m} - A| + |a_n - A| \leq \epsilon$.

□

1.1.6 Series

Definition 1.1.6.1 A **series** is the limit of a formal sum

$$S_n := \sum_{k=0}^n a_k \quad S := \lim_{n \rightarrow \infty} S_n$$

Convergence

Definition 1.1.6.2 Series $\sum_{k=0}^{\infty} a_k$ is **convergent** if the limit exists.

♠ **Proposition 1.1.6.3** For convergent series $\sum_{k=0}^{\infty} a_k, \sum_{k=0}^{\infty} b_k$, we have

$$\begin{aligned} \sum_{k=0}^{\infty} (a_k + b_k) &= \sum_{k=0}^{\infty} a_k + \sum_{k=0}^{\infty} b_k \\ \sum_{k=0}^{\infty} (ca_k) &= c \sum_{k=0}^{\infty} a_k \end{aligned}$$

◦ **Proof** Omitted.

□

♠ **Proposition 1.1.6.4** (Cauchy's Criterion) $\sum_{k=0}^{\infty} a_k$ is convergent iff

$$\forall \epsilon > 0 \exists N (\forall n \geq N \forall p > 0 (|\sum_{k=n}^{n+p} a_k| < \epsilon))$$

◦ **Proof** Omitted.

□

Positive Series

Definition 1.1.6.5 Series $\sum_{k=0}^{\infty} a_k$ is **positive** iff $a_k \geq 0$.

♠ **Proposition 1.1.6.6** Positive series $\sum_{k=0}^{\infty} a_k$ is convergent $\Leftrightarrow \sum_{k=0}^{\infty} a_k \leq M$ for some M .

◦ **Proof** Omitted. □

♠ **Proposition 1.1.6.7** If $a_k \geq b_k \geq 0$, then

$$\sum_{k=0}^{\infty} a_k \text{ is convergent} \Rightarrow \sum_{k=0}^{\infty} b_k \text{ is convergent}$$

◦ **Proof** Omitted. □

Absolute Convergence

Definition 1.1.6.8 $\sum_{k=0}^{\infty} a_k$ is **absolutely convergent** if $\sum_{k=0}^{\infty} |a_k|$ is convergent.

D'Alembert's Criterion

◇ **Lemma 1.1.6.9** (D'Alembert) Suppose $\sum_{k=0}^{\infty} a_k$ is positive series. If there exists some $N \in \mathbb{N}$,

- $r < 1$ such that for all $n \geq N$, $a_n > 0$

$$\frac{a_{n+1}}{a_n} \leq r$$

Then, $\sum_{k=0}^{\infty} a_k$ is convergent.

- such that for all $n \geq N$,

$$\frac{a_{n+1}}{a_n} \geq 1$$

Then, $\sum_{k=0}^{\infty} a_k$ is divergent.

◦ **Proof**

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□

Remark

Cauchy's Root Criterion

◇ **Lemma 1.1.6.10** (Cauchy) Suppose $\sum_{k=0}^{\infty} a_k$ is positive series. If there exists some N ,

- and $r < 1$ such that for all $n \geq N$

$$\sqrt[n]{a_n} < r$$

- such that $n \geq N$

$$\sqrt[n]{a_n} \geq 1$$

Then, $\sum_{k=0}^{\infty} a_k$ is divergent.

◦ **Proof**

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□

Cauchy's

Kummer/Raabe/Bertrand's Criterion

Abel's Criterion

Dirichlet's Criterion

Lebnitz's Criterion

1.1.7 Power Series

Definition 1.1.7.1 A **power series** is a formal sum of

$$f(z) := \sum_{k=0}^{\infty} a_k z^k$$

► **Theorem** (Cauchy-Hadamard) For $\sum_{k=0}^{\infty} a_n x^n$, define

$$\rho := \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{a_n}}$$

Then, the series is absolutely convergent in $|z| < \rho$, divergent in $|z| > \rho$.

◦ **Proof ...**

□

Exponential

Define:

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

$$\exp(x) := \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Geometric Sequence

Definition 1.1.7.2 Geometric Sequence

$$G(z) = \sum_{i=0}^{\infty} z^i = \lim_{n \rightarrow \infty} G_n(z) \quad G_n(z) = \sum_{i=0}^n z^i$$

♠ **Proposition 1.1.7.3**

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- $G(z) = \frac{1}{1-z}, |z| < 1$

◦ **Proof** With $G(z) = \lim_{n \rightarrow \infty} \frac{1-z^{n+1}}{1-z} = \frac{1}{1-z}$. □

Newton's Method**1.1.8 Riemann's Sequence Theorem**▶ **Theorem** (Riemann)

◦ **Proof** ... □

1.1.9 Fubini's Theorem, Tonelli's Theorem▶ **Theorem** (Fubini)

◦ **Proof** ... □

▶ **Theorem** (Tonelli)

◦ **Proof** ... □

1.1.10 L'Hospital's Law**1.1.11 Infinite Product****1.2 Limit of Functions****1.3 Continuous Functions on $\mathbb{R}$**

Definition 1.3.0.1 $f : U \rightarrow \mathbb{R}$ is **continuous** at $a \in U$ if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

1.3.1 Intermediate Value Theorem

◇ **Lemma 1.3.1.1** If $f \in C([a, b], \mathbb{R})$ with $f(c) > 0$, $c \in [a, b]$. Then, there exists $\delta > 0$ such that

$$\forall x \in (c - \delta, c + \delta) \cap [a, b] (f(x) > 0)$$

◦ **Proof ...**

□

► **Theorem** (Intermediate Value Theorem) Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous with

$$f(a) > c$$

◦ **Proof ...**

□

1.3.2 Extreme Value Theorem

Lesbegue Value

1.3.3 title

Chapter 2 Derivative

2.1 Derivatives

Definition 2.1.0.1 Define:

Chapter 3 Integration

3.1 Antiderivative

Definition 3.1.0.1 (Indefinite Function) Suppose

$$f : [a, b] \rightarrow \mathbb{R}$$

Then, $F : [a, b] \rightarrow \mathbb{R}$ is an antiderivative of f if

- F is continuous;
- $F'(x) = f(x), x \in (a, b)$;

◇ **Lemma 3.1.0.2** Suppose F, G be the antiderivative, then we have

$$F - G = C$$

◦ **Proof** Notice that

$$\frac{d}{dx}(F - G) = F' - G' = 0 \Rightarrow F - G = C$$

□

◇ **Lemma 3.1.0.3** (Linearity) Suppose F_1, F_2 be antiderivative of f_1, f_2 , then

$$(\alpha F_1 + \beta F_2 + c)' = \alpha f_1 + \beta f_2$$

◦ **Proof** Omitted

□

◇ **Lemma 3.1.0.4** (Chain's Rule) Suppose $g : D \rightarrow \mathbb{R}, h : I \rightarrow D$ with antiderivative G of g . Then, suppose

$$f = (g \circ h)h' : I \rightarrow \mathbb{R}$$

We have

$$F = G \circ h$$

◦ **Proof** Omitted.

□

3.1.1 Partial Integration

♠ **Proposition 3.1.1.1** ...

◦ **Proof** ...

□

★ Corollary 3.1.1.2 ...

◦ Proof ...

□

3.2 Riemannian Integration

3.2.1 title

3.3 Theorems of Integration in One Variable

3.3.1 Newton-Lebnitz Theorem

► Theorem ...

◦ Proof ...

□

3.3.2 Integrations Intermediate Value Theorem

3.4 Indefinite Integration

3.5 Summe and Integration

◇ Lemma 3.5.0.1 We have: For $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$

$$\sum_{i=1}^n f(n) \text{ is convergent} \Leftrightarrow \int_1^\infty f(x)dx \text{ is convergent}$$

◦ Proof ...

□

Chapter 4 Discusion of Series and Infinite Product, in the Category of Analysis in One Variable

Chapter 5 ODE

5.1 title